CMS Draft Analysis Note

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Cascade: A Python-based Agentic CLI for HPC Clusters

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DRAFT

Abstract

A Python-based agentic CLI for HPC clusters, built on modern harness engineering patterns for HEP. Cascade provides a layered isolation architecture that enables AI-assisted analysis workflows on shared computing infrastructure without contaminating the host environment.

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History of the different versions of the note

- V1: Initial scaffold — chapter structure established with 7 sections.

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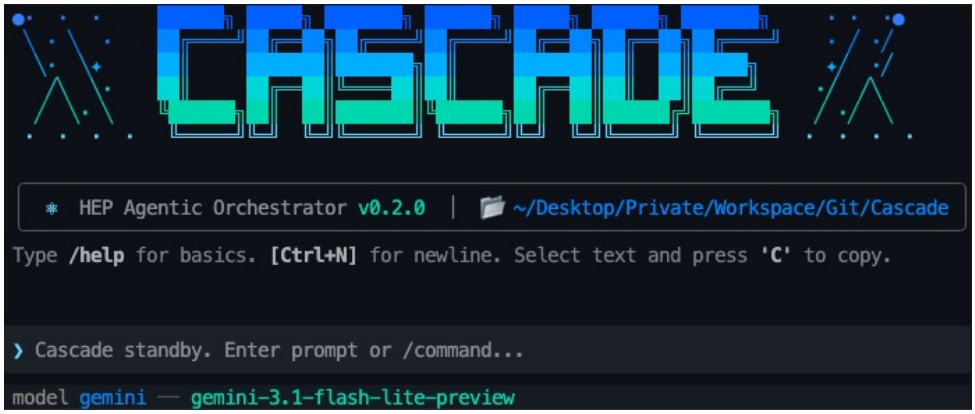

Figure 1: Overview of the Cascade CLI interface — an HPC-native agentic orchestrator for CMS analysis workflows.

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References

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